

Problems for the Elasticity, cantilever geometry (more general formulation + FEM)

Uniform distributed load (f_1, f_2) and boundary load (F_1, F_2) applied at $s=L$

→ the balance of forces can be solved first for the horizontal / vertical components R_1 and R_2 of the contact force $\underline{R}$

$$\underline{R}' + \underline{f} = 0 \quad \left\{ \begin{array}{l} \frac{\partial}{\partial s} R_1 = -f_1 \quad , \quad R_1(L) = F_1 \\ \frac{\partial}{\partial s} R_2 = -f_2 \quad , \quad R_2(L) = F_2 \end{array} \right. \rightarrow \begin{cases} R_1(s) = F_1 + f_1(L-s) \\ R_2(s) = F_2 + f_2(L-s) \end{cases}$$

Plugging $R_1(s)$ and $R_2(s)$ into the balance of moments $\frac{\partial M}{\partial s} = -R_2 \sin \alpha + R_1 \cos \alpha$ we get ($B = \text{const}$)

$$\begin{cases} B \alpha'' = (F_1 + f_1(L-s)) \cos \alpha - (F_2 + f_2(L-s)) \sin \alpha \\ \alpha(0) = 0 \quad (\text{clamp}) \\ M(L) = 0 \rightarrow \alpha'(L) = 0 \end{cases} \quad (1)$$

Euler: $f_1 = f_2 = 0$, $F_1 < 0$, $F_2 = 0$

Galerkin: $f_1 = f_2 = 0$, $F_1 = 0$, $F_2 < 0$

• Finite Element method for (1)

Weak form of (1): multiply by arbitrary $\tilde{\alpha}$ and integrate

$$\int_0^L B \alpha'' \tilde{\alpha} \, ds = \int_0^L (F_1 + f_1(L-s)) \cos \alpha \tilde{\alpha} \, ds - \int_0^L (F_2 + f_2(L-s)) \sin \alpha \tilde{\alpha} \, ds \quad \forall \tilde{\alpha} \in V, \tilde{\alpha}(0) = 0$$

$\alpha \in V = \{ \alpha \text{ is continuous, } \alpha' \text{ piecewise continuous and bounded} \}$

$$\underbrace{B \alpha' \tilde{\alpha}}_{\tilde{\alpha}(0)=0} \Big|_0^L - \int_0^L B \alpha' \tilde{\alpha}' \, ds = \int_0^L (F_1 + f_1(L-s)) \cos \alpha \tilde{\alpha} \, ds - \int_0^L (F_2 + f_2(L-s)) \sin \alpha \tilde{\alpha} \, ds$$

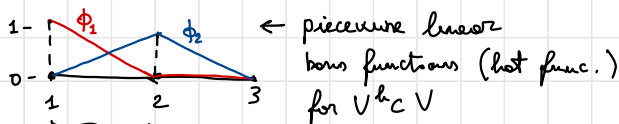
Discretization:

Mesh $i=1$ 0 h L $i=N$ N modes, $h = \frac{L}{N-1}$

$$\alpha(s) \approx \alpha^h(s) = \sum_{i=1}^N \phi_i(s) \alpha_i = \underbrace{\Phi^T(s)}_{(1 \times N)} \underbrace{\underline{\alpha}}_{(N \times 1)} \in V^h$$

row vec column vec

$$\tilde{\alpha}(s) \approx \tilde{\alpha}^h(s) = \sum_{i=1}^N \phi_i(s) \tilde{\alpha}_i = \Phi^T(s) \tilde{\underline{\alpha}} \in V^h$$



Identification of the basis functions ϕ_1 and ϕ_2 :

$$\begin{cases} \alpha^h(s) = \alpha_0 + \alpha_1 s = \phi_1(s) \alpha_1 + \phi_2(s) \alpha_2, & 0 < s < h \\ \alpha^h(0) = \alpha_1 \\ \alpha^h(h) = \alpha_2 \end{cases}$$

$$\Rightarrow \alpha_0 = \alpha_1 \quad \Rightarrow \quad \alpha^h(s) = \underbrace{\left(1 - \frac{s}{h}\right)}_{\phi_1(s)} \alpha_1 + \underbrace{\frac{s}{h}}_{\phi_2(s)} \alpha_2$$

Remark: $\frac{\partial \alpha}{\partial s} \approx \frac{\partial \alpha^h}{\partial s} = \frac{\partial \Phi^T(s)}{\partial s} \underline{\alpha} = \underline{\alpha}^T \frac{\partial \Phi}{\partial s}$ (analogously for $\frac{\partial \tilde{\alpha}}{\partial s}$)

$\frac{\partial \Phi^T}{\partial s}$ is the discrete version of the gradient operator $\frac{\partial}{\partial s}$

Using the discretizations for α and $\tilde{\alpha}$ in the weak form:

$$\underbrace{- \tilde{\underline{\alpha}}^T \int_0^L \Phi^T B(\Phi)^T \underline{\alpha}}_{\substack{K \\ (N \times N, \text{sym})}} = \tilde{\underline{\alpha}}^T \int_0^L \Phi (F_1 + f_1(L-s)) \text{mm} (\Phi^T \underline{\alpha}) - \tilde{\underline{\alpha}}^T \int_0^L \Phi (F_2 + f_2(L-s)) \omega (\Phi^T \underline{\alpha})$$

$\forall \tilde{\alpha}^h \in V^h$

$$\tilde{\underline{\alpha}}^T \underline{K} \underline{\alpha} + \tilde{\underline{\alpha}}^T \underline{F}(\underline{\alpha}) = 0 \quad \forall \tilde{\underline{\alpha}}$$

where

$$\underline{F}(\underline{\alpha}) = \int_0^L \Phi (F_1 + f_1(L-s)) \text{mm} (\Phi^T \underline{\alpha}) - \int_0^L \Phi (F_2 + f_2(L-s)) \omega (\Phi^T \underline{\alpha})$$

Choose $\tilde{\alpha}^1 = [1 \ 0 \ 0 \ \dots \]$, $\tilde{\alpha}^2 = [0 \ 1 \ \dots \]$ to generate N linearly indep. eqs

$\underline{K} \underline{\alpha} + \underline{F}(\underline{\alpha}) = 0 \quad (*)$ system of N non-linear algebraic eqs
(+ prescribed nodal value of α at clamp, i.e. $\alpha_0 = 0$)

Newton method: $\underline{\alpha}^{(k+1)} = \underline{\alpha}^{(k)} + \Delta \underline{\alpha} \rightarrow$ linearize system $(*)$ about $\underline{\alpha}^{(k)}$ ($\Delta \underline{\alpha}$ small)

$\left(\underline{K} + \frac{\partial \underline{F}}{\partial \underline{\alpha}} \right)_{\underline{\alpha}^{(k)}} \Delta \underline{\alpha} = - \underline{K} \underline{\alpha}^{(k)} - \underline{F}(\underline{\alpha}^{(k)}) \rightarrow$ solve for $\Delta \underline{\alpha}$, $\underline{\alpha}^{(k)}$ known, then update
 $\underline{\alpha}^{(0)}$ specified as initial guess

Example: solve numerically Euler's (only 1st buckling load) and Galileo's probs.

• FEM for Euler's problem

To compute critical loads and modes, start from linearized eqn

$$\begin{cases} \alpha'' + \frac{F_1}{B} \alpha = 0 \\ \alpha(0) = 0 \\ \alpha'(L) = 0 \end{cases} \Rightarrow \underline{K} \underline{\alpha} + F_1 \underline{M} \underline{\alpha} = 0 \quad (\text{generalized eigenvalue prob})$$

where $\underline{M} = \int_0^L \phi \phi^T$

Eigenvalues (F_1) $\rightarrow$ critical loads $F_{20}, F_{21}, F_{22} \dots$

Eigenvectors ($\underline{\alpha}$) $\rightarrow$ critical modes

Examples: compute 3 eigenvalues/eigenvectors numerically (consol) and compare with analytical formulas. Compute numerically the bifurcation diagram $(F_1, |\alpha_2|)$ for the first 3 critical loads.

Post-critical branches can be computed by solving numerically the non-linear problem $(*)$

using a continuation approach: use the mode shapes near critical loads as initial guesses for Newton's method.

